
Team Learnova

**Learnova
Software Test Description (STD)**

**Version 1.0
10/31/2025**

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Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within < > are replaced with appropriate text.	YES
The document is prepared in the "Justify Text" format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states "The first version". Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
There are an adequate number of test cases to cover all the necessary testing for the application.	YES
The test traceability matrix includes all the test cases and all application requirements including functional, hardware, and quality.	YES
All test cases in the test case list are detailed in the appendixes.	YES
The planned test duration is adequately estimated for the test case.	YES
All fields in the test cases are completed and no field is left empty.	YES
The test steps in the test case are detailed enough to adequately test the feature/s and the requirements.	YES
The test cases include at least 10 steps that are detailed enough to replicate the test with inputs and outputs.	YES

Final Control	Yes/No
The document is checked based on the "Team Project Artifacts Common	YES

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Errors” Guideline. The document is ready for release.	
The document is not ready for release.	NO

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Document Preparation

Name	Role	Approval (Signature)	Approval Date
Akanksha Bolla	Project Manager	Akanksha Bolla	10/31/25
Dharma Reddy Pandem	Project Design Manager	Dharma Reddy Pandem	10/31/25
Sai Priya Ramini	Project Configuration Manager	Sai Priya Ramini	10/31/25
Bhavan Reddy Karri	Project Quality Assurance Manager	Bhavan Reddy Karri	10/31/25

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Akanksha Bolla	Project Manager	Akanksha Bolla	10/31/25
Keerthana Dallavai	Project Requirements Manager	Keerthana Dallavai	10/31/25
Dharma Reddy Pandem	Project Design Manager	Dharma Reddy Pandem	10/31/25
Sumadhar Rao Vunnam	Project Test Manager	Sumadhar Rao Vunnam	10/31/25
Goutham Darapogu	Project Technical Manager	Goutham Darapogu	10/31/25
Sai Priya Ramini	Project Configuration Manager	Sai Priya Ramini	10/31/25
Bhavan Reddy Karri	Project Quality Assurance Manager	Bhavan Reddy Karri	10/31/25

Revision History

Date	Version	Description	Author
10/31/25	1.0	The First Draft	Akanksha Bolla

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Software Test Description

1. Introduction

1.1 Purpose of the Document

This document explains the testing process planned for the Learnova website. It outlines how the features built so far are tested through well-defined test cases to ensure they work as intended. The document also serves as a practical guide for testers, developers, and other team members to maintain the overall quality and reliability of the system.

1.2 Scope of the Document

This document defines the testing process for Learnova.

It includes:

- Test plans and schedules
- Roles and responsibilities
- Test environment setup
- Test cases and expected results
- Requirement-to-test case mapping

1.3 References

Reference Title/Name	Date	Version
Project Proposal	09/02/2025	1.0
Project Vision Document (PVD)	09/15/2025	1.0
Project Plan (PP)	09/19/2025	1.0
Software Requirements Specifications (SRS)	09/26/2025	1.0
Software Design Description (SDD)	10/31/2025	1.0
IEEE/ISO/IEC ISO/IEC/IEEE 29148-2018 International Standard - Systems and software engineering -- Life cycle processes	10/23/2018	2018
Web Content Guidelines (WCAG) 1.4 Accessibility	11/11/2008	2.0

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Project Description Document	PDD	The project description document is the project document detailing the project to be completed.

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Project Vision Document	PVD	The project vision document is the project document describing the project vision to be accomplished.
Project Plan	PP	The project plan is the project document detailing the plan to be followed throughout the project.
Software Requirements Specification	SRS	The software requirements specification is the project document specifying the project requirements and the use cases to be satisfied for the successful completion of the project.
Software Design Description	SDD	The software design description is the project document specifying the project requirements and the use cases to be satisfied for the successful completion of the project.
Software Test Description	STD	The software test description is the project document specifying the test-related planning, activities, and test cases to successfully verify the software under development.

2. Product Scope

The main scope of Learnova is to generate an AI-powered learning platform that assists students, educators and professionals to process and understand their academic materials. This application enables users to upload documents like PDFs, Word Files, and notes. After processing and understanding the documents, this application generates few learning concepts like summaries, quizzes, flashcards and most importantly chatbot-based Q&A support. The main Features within the scope of the project are that users can upload and organize the study materials, Automatic summarization of lengthy context into easy understanding formats, Interactive flashcards are generated with the help of key concepts, Quizzes are generated by AI to improve the understanding levels and test the learned concepts, For quick clarification of queries user can interact with AI chatbot, A personalized dashboard displaying progress and insights, login is secured and data is protected for each user.

Out of Scope: For now learnova is only a web-based interface but in future mobile application can be added for development, since now it is only accessed with good internet, Enabling voice recognition for chatbot is for future iterations not present in this version, Interface will be made different with advanced analytics for professors or administrators in the later development phases. The main Features within the scope of the project are that users can upload and organize the study materials, Automatic summarization of lengthy context into easy understanding formats.

3. Test Planning

3.1 Responsibilities

Role	Name	Responsibilities
Test Engineer	Bhavan Reddy Karri	The test engineer is responsible for preparing the test cases under the direction of the test manager.
Developer	Dharma Reddy Pandem	The developers support the test engineers during the tests. They fix the defects/bugs if identified during tests. The developers always stay in the loop for the testing process.

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Test Manager	Sumadhar Rao Vunnam	The test manager is responsible for all aspects of the project testing. The test manager plans and coordinates all testing activities, takes corrective action related to the testing activities if needed, and informs the project manager and technical manager regarding the test progress.
Customer/User Representative	Akanksha Bolla	The customer/user representative approves the software test description, reviews test cases, and supports the testing process if possible.
Requirements Manager	Keerthana Dallavai	The requirements manager approves the software test description, reviews test cases, and supports the testing process if possible.
Technical Manager	Goutham Darapogu	The technical manager is responsible for ensuring that all teams (project team members) work in tandem to achieve a successful testing process.

3.2 Test Schedule

Test Case	Scheduled Date	Actual Date	Duration
TC-1	10/20/2025	10/20/2025	1.5 hours
TC-2	10/21/2025	10/22/2025	2 hours
TC-3	10/22/2025	10/23/2025	1 hour
TC-4	10/23/2025	10/24/2025	1.5 hours
TC-5	10/24/2025	10/25/2025	1 hour

4. Test Preparation

4.1 Test Environment

Testing environment in learnova was conducted in a controlled environment to ensure reliable and consistent results. It is deployed on standard computing hardware and for functionality and usage it is used via web browsers.

Environment Details:

Hardware: Standard laptop with at least 8GB RAM, Intel i5, stable internet connection, and no other hardware or equipment needed.

Software & Tools: Windows 10/ macOS for operating system, Google Chrome for browser, in development stack React.js(frontend), Node.js (backend), MongoDB (database). In AI Integration for generating quizzes, flashcards, summarization Hugging Face APIs is used. For testing manual testing is done, there is no usage of automation tools.

Location: Testing was done on individual laptops and verified through multiple devices to ensure compatibility.

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5. Test Cases

5.1 Test Cases

Test Case ID	Test Case Name	Description
TC-1	Login Test	This test scenario confirms that both valid and invalid credentials can be used to log in. It guarantees that users who have registered can log in successfully, and it displays the relevant error messages when credentials are missing or invalid.
TC-2	User Interface Test	This test case checks the dashboard, sidebar, and navigation menus, among other UI elements. Across desktop and mobile platforms, it guarantees uniformity in responsiveness, color, and layout.
TC-3	Summarization Functionality Test	This test confirms that the summarizing module produces clear summaries and correctly analyzes input text. Text-limit validation, error display for invalid input, and proper handling of API responses are all checked.
TC-4	Quiz Generation Test	This test makes sure that AI-based multiple-choice questions are created from user-provided text, hence validating the quiz generator. It verifies the proper answer mapping, question layout, and score feedback.
TC-5	Q&A Assistance Test	The Q&A module, where users submit a question and context, is validated by this test. It guarantees that the system displays confidence levels accurately, highlights supporting text, and generates pertinent responses.
TC-6	Flashcard Creation Test	The accuracy of the front-back pairs produced by the flashcard generator is verified by this test. Answer visibility, responsiveness across devices, and hover/reveal effects are all verified.
TC-7	Profile Management Test	Users' ability to modify their personal data, including name, email, avatar, time zone, and preferences, is confirmed by this test. Data permanence and accurate reflection on later logins are verified.
TC-8	Password Update Test	This test verifies the procedure for changing the password. It guarantees that confirmation mismatches are appropriately handled, the increased password strength is enforced, and the existing password is validated.
TC-9	Sidebar Navigation Test	This test confirms that visitors are correctly directed to the Profile, Flashcards, Quiz, Q&A, and Summarization pages using the sidebar navigation. It looks for seamless transitions and active-link highlighting on desktop and mobile devices.

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TC-10	Responsive Layout Test	This test makes that the layout adjusts to different screen widths without any problems. It confirms that all controls are still reachable on small devices and that the sidebar folds into a mobile drawer.
TC-11	Free-Tier Usage Limit Test	Any key functionality can be accessed up to three times by guest (non-logged-in) users, according to this test. After the fourth try, the system shows an upgrade or login prompt and stops further use.
TC-12	Logged-In Free-User Limit Test	This test confirms that free users who are logged in can complete up to five tasks daily across all modules. The software asks the user to upgrade to a premium plan after they've gone over the limit.
TC-13	Premium Unlimited Access Test	This test confirms that all functionalities are accessible to premium subscribers without any limitations. It guarantees that the premium symbol is shown across the interface and that usage counters are turned off.
TC-14	Stripe Payment and Checkout Test	This test verifies that the Stripe payment gateway integration was successful. It confirms that checkout sessions are generated appropriately, that unsuccessful payments display unambiguous error messages, and that successful payments reroute the user.
TC-15	Premium Upgrade and System Integration Test	The Learnova platform's whole end-to-end workflow, including user authentication, use caps, Stripe billing, and premium activation, is validated by this test case. It confirms automatic upgrade to Premium via webhook, makes sure Stripe payment processing is complete, and verifies access limitations for both logged-in and guest users. Following the upgrade, it confirms that premium status and data are maintained across all modules and sessions and checks for limitless feature access.

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6. Appendix A – Test Case to Requirements Traceability

Test Case ID	Requirement ID
TC-1	Req-Func-Sw-4 Req-Func-Hw-2 Req-Qual-Sec-1
TC-2	Req-Func-Sw-3 Req-Func-Hw-2 Req-Qual-Sec-5
TC-3	Req-Func-Sw-2 Req-Func-Hw-1 Req-Qual-Sec-1

7. Appendix B – Test Case 1 – User Interface Test

Test Case ID	TC-1
Test Case Name	Login Test
Brief Description	This test scenario confirms that both valid and invalid credentials can be used to log in. It guarantees that users who have registered can log in successfully, and it displays the relevant error messages when credentials are missing or invalid.
Priority	High
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025
Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
The version of Software Under Test	1.0

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Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Google Chrome 129.0.6668.71 Visual Studio Code 1.93.0
List of Test Support Hardware	Laptop Dell Windows 11

Test Case 1 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Open the Learnova application and navigate to the login page.	The login page is displayed with fields for Email/Username and Password along with a Sign In button.	Pass
2	Leave both fields blank and click on the Sign In button.	The system displays an error message: “Email and password are required.” No login is performed.	Pass
3	Enter an invalid email and any password and click Sign In.	The system displays an error message: “User not found.”	Pass
4	Enter a valid email and an incorrect password then click Sign In.	The system displays an error message: “Incorrect password. Please try again.”	Pass

8. Appendix C – Test Case 2 – User Interface Test

Test Case ID	TC-2
Test Case Name	User Interface Test
Brief Description	This test case checks the dashboard, sidebar, and navigation menus, among other UI elements. Across desktop and mobile platforms, it guarantees uniformity in responsiveness, color, and layout.
Priority	High
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025

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Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0, Visual Studio Code 1.93.0
List of Test Support Hardware	MacBook Pro (Apple M2, 16GB RAM, 512GB SSD, macOS Sonoma 14.x)

Test Case 2 Scenario

Test Step Number	Input	Expected Outcome	Result (Pass/Fail)
1	Open application on desktop browser	Dashboard, sidebar, and menu render according to guidelines	Pass
2	Resize desktop browser window	UI remains responsive and adapts at breakpoints	Pass
3	Access application via mobile browser	UI matches mobile design specs, elements display correctly	Pass
5	Inspect color scheme and layout	Colors and layouts match brand/design standards	Pass

9. Appendix D – Test Case 3 – Summarization Functionality Test

Test Case ID	TC-3
Test Case Name	Summarization Functionality Test
Brief Description	This test confirms that the summarizing module produces clear summaries and correctly analyzes input text. Text-limit validation, error display for invalid input, and proper handling of API responses are all checked.
Priority	High
Planned Test Duration	1 hour
Current Version	1.0

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Date of First Version	10/18/2024
Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0, Visual Studio Code 1.93.0
List of Test Support Hardware	Laptop Dell Windows 11

Test Case 3 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the summarization application.	The application starts successfully and displays the input interface.	Pass
2	Enter a valid paragraph of 300 words and click Summarize.	The system generates a concise, coherent summary under 100 words.	Pass
3	Enter an empty input field and click Summarize.	The system displays an error message: "Input text cannot be empty."	Pass
4	Enter a very long text exceeding the allowed limit (e.g., 10 000 words).	The system displays: "Text exceeds maximum limit."	Pass

10. Appendix E – Test Case 4 – Quiz Generation Test

Test Case ID	TC-4
Test Case Name	Quiz Generation Test
Brief Description	This test makes sure that AI-based multiple-choice questions are created from user-provided text, hence validating the quiz generator. It verifies the proper answer mapping, question layout, and score feedback.
Priority	High
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025

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Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0, Visual Studio Code 1.93.0
List of Test Support Hardware	MacBook Pro (Apple M2, 16GB RAM, 512GB SSD, macOS Sonoma 14.x)

Test Case 2 Scenario

Test Step Number	Input	Expected Outcome	Result (Pass/Fail)
1	Input a sample text (300+ words) and request quiz generation	System generates multiple-choice questions relevant to the passage	
2	Select an answer for each question	System maps answers correctly and provides instant feedback	
3	Inspect layout and readability of questions	Questions and answers are well formatted and easily readable	
4	Submit completed quiz	System calculates score, displays correct/incorrect responses, and provides score feedback	
5	Input text with ambiguous answers or formatting	System maintains clear answer mapping and prevents option ambiguity	

11. Appendix C – Test Case 5 – Summarization Functionality Test

Field	Details
Test Case ID	TC-5
Test Case Name	Q&A Assistance Test
Brief Description	The Q&A module, where users submit a question and context, is validated by this test. It guarantees that the system displays confidence levels accurately, highlights supporting text, and generates pertinent responses.

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Priority	High
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025
Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
Version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Q&A Assistance Module, API Client Tool (e.g., Postman), Web Browser (Chrome/Edge)
List of Test Support Hardware	Standard Workstation (Windows/Linux)

Test Case 2 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the Learnova application and navigate to the Q&A Assistance module.	The Q&A interface loads successfully and displays input fields for context and question.	Pass
2	Enter a valid paragraph as context and a clear question related to it.	The system processes the input and returns a relevant, accurate answer.	Pass
3	Observe the displayed confidence score for the generated answer.	The system displays a numeric or visual confidence indicator (e.g., 87% confidence).	Pass
4	Submit an unrelated or unclear question with a valid context.	The system provides a fallback message such as “Unable to find relevant answer.”	Pass

12. Appendix C – Test Case 6 – Summarization Functionality Test

Field	Details
Test Case ID	TC-6
Test Case Name	Flashcard Creation Test

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Brief Description	The accuracy of the front-back pairs produced by the flashcard generator is verified by this test. Answer visibility, responsiveness across devices, and hover/reveal effects are all verified.
Priority	High
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025
Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
Version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0, Visual Studio Code 1.93.0
List of Test Support Hardware	Laptop Dell Windows 11

Test Case 2 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the Learnova application and navigate to the Flashcard module.	The Flashcard module loads successfully and displays the flashcard creation interface.	Pass
2	Upload a valid study document or paste educational text.	The system processes the text and shows detected key terms for flashcard creation.	Pass
3	Click Generate Flashcards .	The system generates flashcards showing question (front) and answer (back) pairs.	Pass
4	Hover or tap on a flashcard.	The flashcard flips, revealing the correct answer clearly.	Pass

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13. Appendix B – Test Case 7 – Profile Management Test

Test Case ID	TC-7
Test Case Name	Profile Management Test
Brief Description	This test confirms users' ability to modify their personal data, including name, email, avatar, time zone, and preferences. It verifies that data permanence and accurate reflection on later logins are maintained.
Priority	Medium
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025
Date of Current Version	10/31/2025
Created By	Keerthana Dallavai
Last Update By	Goutham Darapogu
Approved By	Sumadhar Rao Vunnam
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0 Visual Studio Code 1.93.0
Cod	Standard laptop with 8GB RAM, Intel i5 using Chrome browser

Test Case 1 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Open learnova application and navigate to profile page.	Profile page loads successfully. It displays username, email and	Pass

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		preferences.	
2	Edit the email field with invalid address and save it.	The system rejects and gives please enter valid email address.	Pass
3	Upload an unsupported file type and save changes.	The system should reject unsupported file type and then display “invalid file type”	N/A
4	Entering excessively long text in the name field.	The system should restrict the input of the field. Displaying “maximum 20 characters allowed	Pass.

14. Appendix C – Test Case 8 – Password Update Test

Field	Details
Test Case ID	TC-8
Test Case Name	Password Update Test
Brief Description	This test verifies the functionality of the password update feature in the Learnova website. It ensures that users can successfully update their passwords, that validation is properly handled for mismatched or weak passwords, and that secure encryption is applied before saving the new password.
Priority	Medium
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	10/18/2025
Date of Current Version	10/31/2025
Created By	Bhavan Reddy Karri
Last Update By	Bhavan Reddy Karri
Approved By	Sumadhar Rao Vunnam
Version of Software Under Test	1.0
Test Engineer to Conduct the Test	Bhavan Reddy Karri
List of Test Support Software	Postman 11.14.0, Visual Studio Code 1.93.0
List of Test Support Hardware	Laptop Dell Windows 11

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Test Case 8 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the Learnova website and log in with valid user credentials.	The system logs in successfully and displays the dashboard.	Pass
2	Enter the current password incorrectly and submit.	The system displays "Incorrect current password."	Pass
3	Enter a new password and mismatched confirmation password.	The system displays "Passwords do not match."	Pass
4	Enter a valid current password and a strong new password (e.g., <i>Learnova@2025</i>).	The system accepts the update and displays "Password changed successfully."	Pass